

AKASH MAHAJAN

✉ mahajanakash2004@gmail.com 📞 +91 9021497733 🔗 [LinkedIn](#)

EDUCATION

Sardar Patel Institute of Technology

Dec 2022 – May 2026

Bachelor of Technology in Electronics and Telecommunication Engineering

Mumbai, India

- Relevant Course: Database Management Systems

TECHNICAL PROJECTS

PawFinds – Pet Adoption Platform - *React.js, Node.js, Express.js, MongoDB, JWT, OTP*

July 2024

- Engineered a secure MERN stack platform for pet adoption with separate user and admin portals.
- Deployed OTP-based signup/login with JWT authentication and Context API for role-based UI rendering.
- Built an admin dashboard to manage pet postings, adoption requests, approvals/rejections, and adoption history.
- Architected modular React components (PetCards, AdoptForm, Dashboard) and ensured responsive UI with CSS.
- Facilitated complete adoption lifecycle tracking (Pending → Approved → Adopted) with secure CRUD operations.

e-Learning Platform - *MERN Stack, JWT Auth, Redux, Tailwind CSS*

Dec 2023

- Created a full-stack **e-learning system** with separate dashboards for Students, Teachers, and Admins.
- Integrated **JWT-based authentication** with role-based access, enabling secure login for different user types.
- Developed reusable React components (Navbar, Sidebar, Course Cards, Forms) with **React Router** for navigation.
- Orchestrated **course enrollment, content upload, and payment flow**, allowing real-time updates via Redux state management.
- Crafted a responsive and scalable UI with **Tailwind CSS**, enhancing user experience across all devices.

Marsupial Robotic System – Autonomous UGV–UAV with LiDAR Mapping & Precision Landing April 2025

- Conceptualized a marsupial robotic system where a UGV performs LiDAR-based SLAM and serves as a mobile landing platform for UAV deployment and recovery.
- Implemented **SLAM (Cartographer/LIO-SAM)** and **path planning (Nav2)** for navigation with obstacle detection in complex indoor environments.
- Integrated **AprilTag detection** with a downward UAV camera for autonomous precision landing, supported by a **PID/visual servoing controller** with ROS 2–PX4 coordination.
- Enabled extended exploration, safe recovery, and multi-robot coordination for **search-and-rescue, industrial inspection, and planetary exploration**.

POSITIONS OF RESPONSIBILITY

Technical Head - *Enactus S.P.I.T*

2024

- Led the creation and maintenance of the official club website, ensuring functionality, usability, and regular updates.
- Supported organizational activities by delivering technical solutions and infrastructure for events and initiatives.
- Mentored students in technical innovation, fostering skills in web development, problem-solving, and project execution.

Contingent Leader - *Mood Indigo I.I.T Bombay*

Dec 2023

- Led and coordinated the team for Mood Indigo events, improving productivity and engagement.

Indigo Squad Member - *Mood Indigo I.I.T Bombay*

Dec 2023

- Assisted in planning, coordination, and on-ground execution of events.

KEY ACHIEVEMENTS

Design Patent – UGV Rover with LiDAR (United Kingdom)

2025

- Secured a registered design patent for an innovative UGV rover equipped with LiDAR for autonomous mapping and navigation.
- Recognized for applying novel mechanical and robotic design principles to enable real-world exploration and rescue applications.

Gen AI Foundations Certificate - in upGRAD in collaborating with Microsoft.

July 2025

- Completed Generative AI Foundations (upGrad Microsoft), gaining hands-on skills in prompt engineering, content creation, automation, and applying AI to marketing, storytelling, and research.

PROGRAMMING SKILLS AND ANALYTICAL SKILLS

Languages: Python, Java, C++

Databases: MongoDB, MySQL, PostgreSQL

Backend: Node.js, Express.js

APIs: REST APIs, OpenAI

Analytics: PowerBI, Excel

Tools: Postman, VS Code, Figma